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DESIGN OF RECONFIGURABLE CACHE MEMORY WITH INTEGRATED MBE-DCC

A PROJECT REPORT

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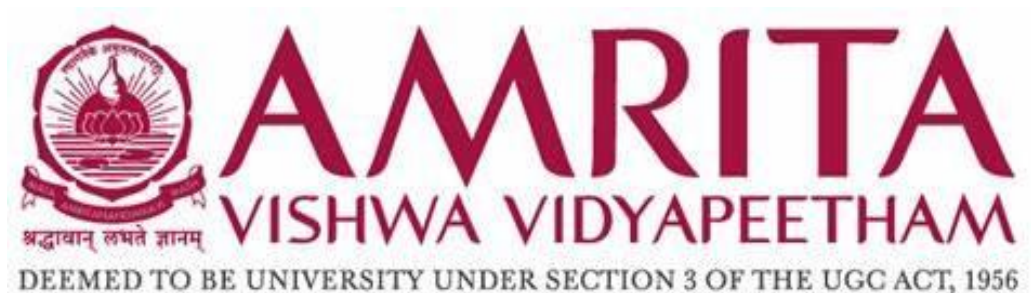
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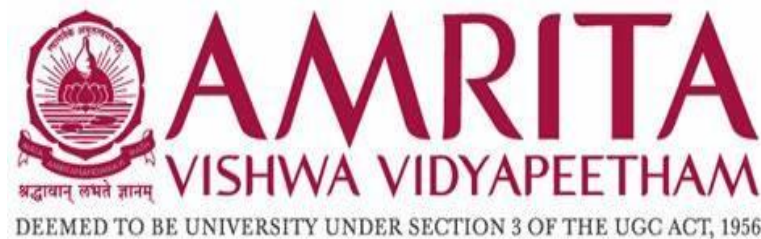
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BONAFIDE CERTIFICATE

This is to certify that the project report entitled “**DESIGN OF RECONFIGURABLE CACHE MEMORY WITH INTEGRATED MBE-DCC**” submitted by

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EXAMINER 1

EXAMINER 2

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Abstract

Modern processor cache memories are being affected by radiation effects, voltage fluctuations and process-induced noises, leading to a higher tendency to adjacent multi-bit burst errors. Typical SEC-DED codes can't be used to correct burst faults and to turn on stronger ECC protection at all time will lead to excessive area and power consumption. The proposed work presents a Reconfigurable Cache Memory along with Adaptive Multi-Bit Burst Error Detection and Correction (MBE-DCC) circuits. It implemented a 2-way set-associative cache memory in synthesizable Verilog HDL with on-chip BCH-based burst error correction encoders and decoders capable to tolerate both 4-burst and 6-burst error correction and detection in the same hardware architecture

The adaptive ECC Mode Manager circuit autonomously switch on either 4-burst mode or 6-burst mode according to the faults happening during run time to obtain less switching activity, delay and power consumption while running in lightweight 4-burst mode under normal condition and more efficient fault-tolerant at abnormal condition of running in 6-burst mode after encountering three consecutive uncorrectable errors. As long as it enters into stable condition where it has some successive minor correction, the architecture will switch to lighter mode again. Four kinds of ECC configuration (4-Burst Only, 6-Burst Only, Adaptive 6B+4B, and Adaptive 4B+3B architectures) were synthesized and simulated using Cadence Genus 21.14 in GPDK045 45nm standard-cell library under slow-corner (1.08V, 125C, 10ns clock) environment. All of the proposed design solutions passed the timing constraints. No violation and negative total slack is seen while running simulation and synthesizing. Static synthesis overhead of adaptive architecture is larger, mainly due to the double decoder and mode control circuit. The gain in the run time compensates for it. Adaptive 6B+4B architecture had a best trade-off between performance, robustness and adaptive feature considering real-time switching.

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List of Abbreviations and Nomenclatures

Abbreviation	Full Form
PCM	Parity Check Matrix(error-correcting code family)
ECC	Error-Correcting Code
MBE-DCC	Multi-Bit Burst Error-Detection and Correction Code
SECDED	Single-Error Correcting, Double-Error Detecting
RTL	Register-Transfer Level
HDL	Hardware Description Language
DXA	Device-independent units (1440 DXA = 1 inch, Word/OOXML unit)
VLSI	Very Large Scale Integration
GPDK045	Generic Process Design Kit for 45 nm technology node
MUX	Multiplexer
FF	Flip-Flop
SDFF	Scan-enabled D Flip-Flop
TNS	Total Negative Slack
QOR	Quality of Results
SDF	Standard Delay Format
SDC	Synopsys Design Constraints
VCD	Value Change Dump (switching activity annotation format)

H-matrix	Parity-Check Matrix (used in cyclic/linear block code design)
XOR	Exclusive-OR logic gate
XNOR	Exclusive-NOR logic gate
NAND	NOT-AND logic gate
NOR	NOT-OR logic gate
AOI	AND-OR-Invert complex gate
OAI	OR-AND-Invert complex gate
LUT	Look-Up Table (FPGA resource)
L1/L2/L3	Cache hierarchy levels (Level 1, 2, 3)
SEU	Single-Event Upset (radiation-induced bit flip)
MBU	Multi-Bit Upset
SRAM	Static Random-Access Memory
FPGA	Field-Programmable Gate Array

CHAPTER 1: INTRODUCTION

1.1 Overall System and Background

On modern processors cache memories are at the highest level of memory hierarchy and is the most critical on-chip design. Cache memories are also the highest performing, most dependable of the critical on-chip designs. L1 cache runs at the processor clock and provides instructions and data to the execution units. L2 cache is a middle man to the L1 cache and the last-level cache (LLC) which is very large. The LLC cache is sometimes referred to as L3 or L>3 and it is shared among the numerous cores on a processor. The LLC is the largest cache on chip with size as little as few megabytes on mobile processors up to tens of megabytes on server chips. All the data from various active threads and various processors are placed in this LLC, and thus the cache has to be available for the entire processor operation and therefore it has the largest window of exposure time to radiation and noise.

In silicon the bit flip is attributed to a physical mechanism. The Single Event Upset (SEU) causes a bit flip in a transistor because a neutron or alpha particle impacts on the silicon lattice of the transistor creating an electron-hole pair which flips the bit. If neighboring bits flip on close proximity cells then this is termed a Multi-Bit Upset (MBU). Due to the scaling down of cache cells down to 45nm process technology, each cell is very close to many other cells. When one particle strikes a cache line then 3-6 cells flip on successive and can become a burst error. Most modern processors use SECDED codes on cache memories to rectify these problems and thus can fix only one bit per codeword but can detect two bits per codeword. Therefore, SECDED codes cannot be used to fix 3 or more consecutive bit flips which forms a burst error.

Using a stronger 6-burst PCM ECC code on top of a 4-burst one is not power efficient for practical purposes because multi-bit bursts would be infrequent enough most of the system lifetime. The 6-burst BCH code has a larger number of parity bits, a deeper XOR syndrome computation tree and a larger correction decision network compared to a 4-burst code, hence having a longer critical path, thus a smaller clock speed and higher switching power per read. Therefore, even if system has to tolerate 6-bit bursts, although 4-bit (or less) burst errors occur most of the time, using the stronger 6-bit decoder every single time is neither energy and time efficient.

The goal of this project is to overcome the problem by designing a configurable cache with adaptable MBE-DCC circuit, which contains both 4-bit and 6-bit PCM decoders, and also an intelligent ECC Mode Manager to observe runtime error events and automatically select the 4-bit or 6-bit decoder on-the-fly based on current error rate. If the operating environment is relatively safe and error rate low (most of the time during system life time), the 4-bit decoder would be utilized in order to have a smaller critical path and reduce the switching power; if the operating environment is increasingly risky (3 uncorrectable error events would trigger the Mode Manager to switch to the stronger 6-bit decoder; if the error rate goes back to normal again, the decoder will revert to 4-bit mode).

A physical design of the 2-way set associative cache which has 6-bit address width, 4 sets, 2 ways per set, 16-bit data width and 7-bit parity extension, therefore total codeword size of 23 bits, are illustrated here. All four configurations are implemented in synthesizable Verilog HDL and synthesized in Cadence Genus 21.14 using GPD045 45 nm standard cell library under slow corner condition with supply voltage 1.08V and junction temperature 125C. All four configurations met timing closure with zero violation under 10 ns clock period.

1.2 Objectives

Main Objective:

The primary goal of the project is to design and implement reconfigurable cache memory architecture along with adaptive multi-bit error detection and correction codes to make memory reliable in presence of burst error and at the same time keeping the hardware overhead and performance low.

Sub-Objectives:

- Design and simulate a 2-way set associative cache memory architecture which includes tag memory, data memory, valid bits and hit/miss detection logic.
- Implement the 4-burst and 6-burst MBE-DCC (Multi-Bit Error Detection and Correction) encoder and decoder architectures to detect and correct adjacent multi-bit errors in the cache data.
- Design an adaptive ECC mode management system which allows to switch ECC modes at runtime based on the current detected error pattern.
- Realize the design architecture in Verilog HDL adopting the synthesizable design flow.
- Simulate the cache functionality and the error detection and correction behavior with error injection technique.
- Synthesize the designed hardware using Cadence Genus and measure the hardware performances including area, power and timing constraints.
- Compare the obtained results with other different ECC modes and evaluate the tradeoff between fault tolerance and hardware cost.

1.3 Methodology

The design work of this project is divided into six phases, so as to suitably design, implement, test and analyse the proposed cache memory architecture.

Stage 1 - Literature survey and architecture selection:

Literature survey on burst error ECC schemes, PCM code theory and cache memory architecture is carried out. The H-matrix structures of the 4-burst and 6-burst PCM code are found by referring from the interleaving based parity overhead minimized architecture presented by Li et al. (2018). The two-way set-associative cache architecture is chosen based on the trade-off between the hardware complexity and the miss penalty at conflict misses.

Stage 2 - Architecture Design:

This phase concentrates on the architectural design of the cache memory system. A two-way set-associative cache configuration along with the design of tag array, data array, valid bits and hit/miss logic is designed. Data paths for interleaving of MBE-DCC encoder and decoder with read and write operation is done. The adaptive mode switching mechanism is designed carefully so as to support both 4-burst and 6-burst error correction.

Stage 3 - Verilog HDL Implementation:

Once the architecture is finalized, the entire cache memory system is designed in Verilog HDL. Modules are designed individually for Cache controller, syndrome generator, ECC encoder, ECC decoder, adaptive mode manager and burst error correction logic. The entire design is coded in a synthesizable form.

Stage 4 - Functional Simulation:

After the completion of the HDL design, the design is verified by simulating its functional behaviour. Different read and write operation on the cache memory along with hit and miss are tested. Cache lines are loaded with a line at the cache way and different burst errors are injected to test the syndrome generator, correction logic and the adaptive mode switching.

Stage 5 - Synthesis and Hardware analysis:

The design is synthesized using Cadence Genus in the GPDK045 standard cell libraries and the hardware parameters, such as area consumed, power dissipated, timing delay, and setup slack were estimated.

Stage 6 - Comparative analysis:

The results from the simulation and synthesis phase are analysed. The different configuration of the ECC system is compared based on the advantages and disadvantages, such as power consumed, reliability, and error correction capability.

1.4 Expected and Obtained Results

The overall assumption made was that the adaptive configuration should have marginally higher area and power than each single mode baseline (as both decoders are to remain on the chip), but not necessarily the reverse (in the run-time environment) (as the non-selected decoder will not be accessed) which was confirmed by the synthesis results.

Config 0 (4-burst only) had by far the lowest area (4606.34m²), power (650.96W) and highest maximum frequency (200.6MHz) as expected due to its simplicity.

Config 1 (6-burst only) has a marginal 3.3% increase in area and 5.0% increase in power from Config 0 implying that the cost of single mode protection against a 6-burst error in silicon area is not large.

Config 2, has a 36.0% increase in area and 37.5% power overhead from Config 0, this reflects the cost in silicon area of having both decoders present and ready to switch. Config 3 has only 4.3% lower area and 4.5% lower power than Config

2 implying the smaller cost of the 3-burst H-matrix than the 6-burst equivalent in silicon area.

No timing violations or negative overall slack was encountered for any configurations at the slow corner. The fact that no negative slack was found implies all the designs should be manufacturable at the intended technology node. As expected all designs show that their respective critical paths lie entirely in the mode manager register, not in any part of the ECC circuitry.

1.5 Timeline

Period	Work Completed
November 2025 – January 2026	Literature Survey on Burst Error ECC, PCM Codes, cache memory. Identify H-matrix structure for 4-burst, 6-burst error correction. Design and implement the ECC encoder and decoder module in RTL for Phase 1.
February 2026	Complete Phase 1. Design and simulate standalone 4-burst, 6-burst ECC encoder/decoder modules in RTL. Verify syndrome generation and burst correction block.
March 2026	Initiate Phase 2. Design a 2-way set-associative cache model. Integrate the ECC encoder and decoder modules into the cache's write and read path respectively. Design hit/miss detection block, tag memory and valid bit.

April 2026	Design adaptive mode manager. Implement ECC Mode Manager in RTL and a counter for tracking uncorrectable errors and a burst size monitor and use these to manage MUX selection to the decoder. Test and simulate controller in RTL by injecting errors.
May 2026	Synthesize all four configurations using Cadence Genus 21.14 and GPDK045 45nm slow corner library and report Area, power, and timing for the four configurations. Draw a final comparison report.

Table 1.1 Timeline**1.6 Organization of the Report**

Chapter 2 describes the State of the art in cache Burst-Error Correction and literature associated with PCM codes, Reconfigurable ECC system and soft error is analysed. Reconfigurable cache system is presented with architecture and design where each sub module and interconnections are described in Chapter 3. The system is implemented in Verilog RTL, testbench design and test results (including waveforms and injected errors) are elaborated in Chapter 4. Cadence Genus is used to obtain full synthesis results (time, area, power, gate details) for all the four configurations in Chapter 5. QOR and comparison of the design with others is finalized in Chapter 6 and 7 respectively and future work is addressed

CHAPTER 2: STATE OF ART

2.1 Introduction

The purpose of having cache memory inside the modern processor to improve the performance due to large main memory access times (long delays, very long). High speed of the cache memory combined with high frequency of the cache memory accesses by the processor result in many difficulties with in-order updating of the cache memory. Because the semiconductor technology is becoming more and more compact, SRAM based cache memories become more and more susceptible to soft errors, radiation, voltage noise and variations to the manufacturing process. All of these problems have the effect to increase error rates and cause problems in the overall operation of the processor.

Older memories mainly suffered single bit errors, and error correction methods such as Parity or SECDED were well sufficient to cope with these. Modern highly scaled memories, in general SRAM's are very prone to multi-bit errors (which can be the result of radiation or multiple cell upset) that conventional ECC's do not always support efficiently (especially on modern devices)..

In order to have a reliable system, various error correction architectures have been developed: PCM codes, Reed-Solomon codes, LDPC codes and Multi-Bit Error Detection and Correction codes (MBE-DCC). Among these approaches, the syndrome based burst error correction techniques are more useful in term of error correction capability and less complex in hardware structure which is useful for cache implementation. Also, many recent approaches focus on adaptive ECC mechanism, where the error correction strength of cache system is adaptively adjust based on the number of detected errors. Adaptive systems can provide higher protection against catastrophic burst error and do not incur penalty on power consumption and area in normal operation.

One other issue which has becoming an important factor to consider for designing of modern caches is the tradeoff between performance and reliability. High error correcting schemes always implement more number of parity bits and more complex syndrome generation circuit and decoding logic resulting in higher hardware cost (area), higher power consumption and also increase the timing delay. Therefore for a high error correct cache design, more stress is given to maximize the error correction ability of the cache system with a minimal overhead of the hardware implementation. Hence recent works try to achieve higher error correction strength, and minimum delay, less switching power and also try to get an adaptive system.

Basic of cache memory, set associative organizations, burst error, syndrome decoding and PCM based ECC techniques will be introduced in this chapter. At the end, all the available literature on adaptive cache reliability architecture and burst error correction scheme will be presented so that the necessary knowledge can be built up for the design of reconfigurable cache memory with adaptive MBE-DCC.

2.2 Basic Background

2.2.1 Cache Memory Organization

Cache memory is a small fast memory placed between processor and main memory which is helpful to reduce access time and increase speed. Processor seeks for the required data from cache memory, it would be searched for only when not found from slow main memory. When it is found it would be accessed from cache memory at high speed and reduces the execution time. If required data is found in cache memory is called cache hit and if found from main memory it is called cache miss.

Present day, every modern processor has multi-levels of cache memory L1, L2, L3 cache memory. L1 cache is the smallest and fastest cache and closest to the

processor core whereas L2 and L3 are of large memory size with lower speed.

Cache memory can be made of SRAM as it accesses the memory at higher speed.

Structure of Cache memory relies upon how the blocks of main memory are mapped to locations within cache. 3 widely used cache mapping techniques are direct mapped cache, fully associative cache and set associative cache. Among the above techniques, set associative cache has emerged as a successful combination of hardware simplicity and high cache hit rate. Cache holds the most frequently used instruction and data blocks and therefore, corruption of data within cache will adversely affect the performance of processor. Hence, the reliability of the cache becomes a paramount factor in the design of modern computer systems.

Scaling causes a greater number of memory elements to be packed into a unit volume in cache and hence they become more prone to radiation induced soft errors and multi bit upsets. Multi bit upsets lead to faulty computation as data corruption causes a misrepresentation of information. Hence it is conventional practice in the design of cache to include ECC (Error Correction Code) to detect and correct read operation errors.

2.2.2 Set Associative Cache Memory

It is superior to direct mapped cache and on the other hand much lower complex than fully associative organization of cache. Set associative organization of cache means that the memory is partitioned into sets and each set is responsible for storing more than one cache line which is also called way. Each block of main memory can be mapped into any of the way in the set.

When the cache is accessed, a given memory address will contain 3 parts: tag bits, index bits and offset bits. The index bits is to distinguish which set the given memory block belongs to and the tag bits is to match the given block with cached blocks by comparing it with each cached block's tag. When they are matched, a

cache hit is encountered (valid bit is also set). Cache miss will happen when the cache block is not found in the memory or valid bit is reset.

Set associative organization of cache provides more cache hit rate and has less conflict miss than direct mapped cache. The complexity of hardware increases due to the several tag comparisons. The two-way set associative cache organization is a very commonly adopted scheme for modern processor systems, for it has a good trade-off on the hardware area and power consuming between itself and four way set associative cache organization.

Two way set associative cache organization along with an integrated adaptive ECC mechanism is adopted in this work. Data stored in the cache are ECC-protected codewords which can be retrieved correctly in spite of possible burst errors occurring.

2.2.3 Error Correcting Codes (ECC)

ECCs are used widely in digital systems to provide integrity of data in storage and transmission. The concept of ECCs is to introduce redundancy to the original data before writing it to the memory in the form of parity bits.

When reading the data back from the memory, parity information is verified against the read data, to detect any corruption to the data. Parity check is one of the simplest ECCs that append one parity bit to the original data, just to detect one bit error, but parity check cannot correct the error and it cannot satisfy modern memory system.

ECCs which can detect double bit errors and correct one bit error are named SECDED, this type of ECC has been applied widely in previous cache memories because the minimal area overhead and latency penalty. With increasing cell density in memory devices, there will be higher chance to corrupt successive bit

cells, so stronger ECCs such as PCM code and MBE-DCC are designed to tolerate burst errors.

The generation of parity bits in PCM code uses generator polynomial and parity check matrix, the corrupted bits are detected by checking the syndrome value of the received codeword. But strong ECCs will bring significant area, power and timing penalty, so adapt ECC structures are proposed in order to make the strength of ECCs adapt dynamic to the circumstances.

2.2.4 Burst Errors in SRAM Memories

A burst error is any event that corrupts sequential bits in a memory system. Today, there are numerous factors that can cause burst errors in modern day SRAM-based cache memories such as radiation particle strikes, power supply voltage fluctuations, manufacturing process variations and age related effects in the transistors. It is known that as transistors become smaller that one disturbance can effect multiple surrounding cells within the memory simultaneously.

Standard SECDED ECC's are efficient at handling single bit errors; however, multiple cell upset corrupts a multiple of neighboring bits within the memory, thus causing a burst error that cannot be detected by SECDED ECC's. Undetected/uncorrected burst errors will dramatically reduce the reliability of the processor due to execution instructions and/or data flowing throughout the processor system becoming corrupt.

Cache memories are highly susceptible to burst errors since cache memories have high usage and maintain current copies of valuable data used by the processor. Data within a cache being corrupted and/or detected could cause the system to incorrectly execute a program, or in worst case scenarios fail completely. Thus, there is a need to devise a high performance cache ECC system for the detection and correction of burst errors which imposes low

hardware overhead overhead penalties, yet can correct multiple consecutive failures.

Several ECC schemes are currently in place that are based on information which can resolve burst error problem with minimal hardware overhead such as BCH-based MBE-DCC; adaptive ECC extends this to provide more correction when such is most likely needed.

2.2.5 Syndrome Calculation and Decoding

The syndrome calculation is the most vital operation in an error correction system since it detects whether there are errors present in the received codeword or not. In encoding process, the parity bits are calculated using the parity check matrices and are appended with original data. In decoding process, received codeword is multiplied with the parity check matrix and the syndrome vector is generated:

If syndrome is zero then the codeword received is error-free. If it is a non-zero syndrome value then the bits in the codeword are corrupt. Every syndrome has a specific bit that is incorrect. The decoder checks syndrome value, matches it with syndrome-error relation and decides error position.

In case of a burst error correction system syndrome calculation becomes complex, as it can involve more number of consecutive incorrect bits at a single time. Not only single-bit errors but multi-bit bursts are to be detected. The more enhanced, syndrome-based MBE-DCC structures are developed with intelligent parity-check matrix calculations, lookup-tables and thus perform burst correction with a very less complexity.

The work proposed in this paper used syndrome-based decoding scheme for the correction of both 4-burst and 6-burst errors. The decoder first switches to lightweight 4-burst logic and if it finds that the error it detected is greater than

that of the existing correction mode, it dynamically changes the mode to the greater correction power mode of 6-burst correction.

2.2.6 BCH-Based Multi-Bit Error Correction Codes

Acyclic error correcting codes Parity Check Matrix (PCM) codes is a better choice to be applied in digital communication and memory system application due to their strong power of detecting and correcting multiple random and burst errors. According to generation polynomials over finite fields, PCM codes can obtain various error correcting capabilities by varying the parity bits.

In the encoding part, the data bits are multiplied by generation matrices to obtain the parity bits which will be added to the data bits, the whole transmitted bit string is a codeword. When decoding, the receiver applies the syndrome check of received codeword to examine if the errors are presented. With the error location algorithm, the error position can be determined and hence error bits will be corrected.

As an error detection and correction codes, there are many sorts of error detection and correction codes among which, BCH based Multi-Bit Error Detection and Correction codes (MBE-DCC) is very well suited for SRAM cache memory due to its well-balanced and excellent capabilities on the correction of adjacency burst errors. While the performance of a single error correction code (ECC) on the basic level compared with PCM codes shows that the error correcting capability of PCM codes is far better, its overheads of reasonable amount are required and the parity bits number and complexity are greater to obtain high error correcting power.

In this proposed work, the architecture based on PCM-inspired adaptive MBE-DCC scheme is designed and implemented. The suggested architecture can offer the ability of 4-burst and 6-burst error correction. At normal operations, the

entire system would be more reliable without the overhead induced by worst-case setting via an adaptive mechanism.

2.3 Related Literature in Detail

2.3.1 Improving Cache Power Efficiency with an Asymmetric Set-Associative Cache

Another approach which uses asymmetric set-associative organization for cache power efficiency without unacceptable sacrifice of the processor performance was proposed by Zhigang Hu et al. The authors stated that while a higher cache associativity can improve cache hit rate significantly, it consumes too much power due to the frequent switching of multiple ways when performing a memory access.

They introduced an asymmetric cache organization, in which different cache ways are operating under different conditions to improve cache performance. The amount of the cache memory being utilized during a memory access is selected to be appropriate to improve cache power efficiency by decreasing the switching activity.

The simulation results show that cache power efficiency was improved significantly without sacrificing the cache power efficiency by the implementation of an asymmetric organization. Optimizing cache organization seems to have potential in hardware efficiency and power performance trade-off. A disadvantage of this approach, according to the authors, is the necessity of complex cache controller to implement the mechanism of different ways operation and selective access. The reliability issues of the cache system are not discussed.

This paper is useful as an reference to our project on adaptive cache architecture with hardware efficiency and reliability and could shed light on the cache organization for the current design.

2.3.2 Reliability Trade-offs in Volatile and Non-volatile Cache Design

In [8], M. **Vinodhini** et al. Designed a syndrome based multi-bit error detection and correction architecture for memory reliability by means of generator matrices and parity-check matrices to prevent adjacent multi-bit failures. Authors states that conventional ECC such as SECDED is not sufficient for burst error correction for advanced scaling memories in SRAM.

This architecture encodes data by calculating parity bits using generator matrices and syndromes using parity-check matrices during decoding process. Syndromes detect position of errors among bits in the received code word. Syndrome patterns are analysed to locate and correct multiple failures. Results revealed enhanced burst error correction capability and stable multi-bit fault detection capability over conventional single-bit ECC; it corrected multiple adjacent errors while the overhead is well acceptable.

On the other hand, to gain more fault correction capability we need to add more parity bits and more complex syndrome decoding circuitry that introduces area overhead and timing delay. This literature is of utmost importance to the presented work because the proposed adaptive cache memory uses syndrome-based decoding, parity-check matrices, and multi-bit burst correction schemes.

2.3.3 Multi-Bit Error Detection and Correction Codes Implementation Using Generator-Parity Check Matrix Technique

G. Shanthi et al. Presented a matrix based multi-bit error detection and correction scheme so as to improve reliability for SRAM based memory system. They have considered enhancement in occurrence of adjacent burst error caused by radiation effect and multi-cell upset in current technology.

Generator matrices used for encoding, and parity-check matrices used for syndrome generation and decoding are utilized in the design proposed. When fetching data, syndrome values are obtained to verify whether the burst errors occurred or not. Through decoding of the syndrome pattern, the decoder will determine the locations of faulty bits and correct it.

Experimental results illustrate that the presented matrix-based correction scheme could successfully tolerate multiple adjacent error and tolerate the fault. It has been illustrated that syndrome-based decoding is an efficient approach for correcting burst error with the relatively small hardware overhead as opposed to the some tough traditional ECC. Nonetheless, additional correction power will lead to a larger generation and decoding circuit, increased timing delay and increased hardware area, as a result of increased parity bits and correction circuitry.

This work can support the present project since the presented adaptive MBE-DCC cache architecture also adopts matrix-based syndrome decoding to correct the burst error within the cache memories.

2.3.4 High-Performance and Flexible Design Scheme with ECC Protection in Cache Memory

Y. Zhou et al. Proposes a high performance flexible ECC protection for cache memory systems. The goal of this design is to enhance reliability, but eliminate useless hardware overhead. They find out that typical static ECC architectures are always working with maximum correction ability. Which increases power consumption and timing delay even in normal working situation. To solve this problem, they designed a flexible ECC architecture which can dynamically switch its correction ability according to the needs of reliability.

The authors use programmable syndrome generation circuit, refined pipeline stages and dynamic correction circuits to allow cache memory working in normal condition without redundant hardware and logic overhead. The test result shows that this design achieves more cache reliability and better timing performance than the static ECC systems.

The flexible ECC mechanism can save meaningless correction overhead and provides powerful protection only when extreme error condition occurs. One weakness of the design is the complex control logic is needed to perform the switching between different modes and adaptive correction. Moreover, more hardware is used to manage flexible configuration logic. This paper is highly relevant to the project, since their idea of dynamic switching between different correction levels is the origin for my project of the adaptive 4-burst and 6-burst ECC architecture design.

2.3.5 Modelling Energy and Delay for ECC-Enabled Cache Architectures

D. Joshi et al introduced analytical models to investigate how ECC mechanism can affect cache memory power and timing performance. The authors indicated that stronger ECC techniques would lead to better reliability at the cost of increased power dissipation, hardware complexity, and critical path delay in cache systems.

The introduced work provided analysis of different ECC-enabled cache designs, estimates contribution of parity generation, syndrome calculation circuits, and decoding hardware to cache power and timing delay. Various ECC techniques were studied to provide some analysis on the trade-off between ECC reliability and hardware overhead.

From their simulation, we found that strong ECC schemes consume much more energy and introduce longer timing delay due to their complex syndrome decoding and correction mechanisms. It stressed the significance of optimizing ECC hardware design to achieve balance between reliability and performance.

One constraint of the proposed work is that it was mostly an analytic modelling and lacked adaptive switching. The proposed work also did not focus on burst-error adaptive correction system.

The proposed work has its relevancy toward the research on the proposed adaptive ECC architecture, since the latter also aims to reduce power and timing overhead in normal operation condition and provide stronger burst correction capacity when necessary.

2.3.6 MCU Tolerance in SRAMs Through Low-Redundancy Triple Adjacent Error Correction

An L.J. Saiz-Adalid et al.[5] suggested a triple adjacent low-redundancy error correction architecture for reliable SRAM based multi-cell upsets. In the mentioned work, it was stated that advances in SRAM technology makes them more vulnerable to radiation-induced adjacent bit failures which cannot be efficiently corrected by the traditional SECDED techniques.

A low-redundancy triple adjacent error correction architecture using adjacency-aware syndrome decoding mechanisms is proposed which corrects adjacent burst errors with minimum parity overhead by addressing improvement in correction ability with less redundant memory elements and hardware overhead. They showed the improved capability of correcting the adjacent multi-cell upsets and burst errors as compared to traditional ECC techniques and simultaneously minimizing the redundancy requirement with adequate reliability.

However, one problem is identified which is that increasing capability to correct the adjacent multi-cell upsets require higher complexity in syndrome decoding logic and correction hardware, so expanding the burst pattern also increases complexity of implementation. This literature is most relevant to the proposed work as cache architecture proposed in this paper will also try to provide adjacent burst error correction for the SRAM based cache memory using adaptable burst correction logic.

2.3.7 MCU Tolerance in SRAMs Through Low-Redundancy Triple Adjacent Error Correction

Michael A. Bender et al. Proposed a detailed study on relating cache associativity and cache performance. It is essential in modern set-associative cache designs to determine whether to use higher levels of associativity which

have the potential to gain the cache hit rate but at the cost of the hardware complexity and memory access time.

They analysed the gains achieved by the various associativity levels as a function of miss rate, processor performance and hardware complexity. An optimum level of associativity has been suggested, beyond which there is an insignificant improvement in the cache performance even after an increment in associativity. It showed moderate level of associativity achieved the optimal trade-off between the cache performance and hardware cost. The cache hit rate increased exponentially and conflict miss rates reduced at a faster rate while the increase in cache performance became slow after a particular associativity level with higher penalty to the tag comparison cost, power consumption and memory access time.

One of the constraints observed is that the literature concentrates only on the structure of the cache, memory access times and performances but doesn't address on memory reliability issues and ECC integration. The proposed paper does not address on any adaptive ECC for burst errors correction.

The literature discussed above appears to be appropriate to the topic of this project as it describes a 2-way set associative cache structure which aims to overcome the hardware complexity vs performance drawback, and integrate a reliability factor of adaptive ECC

2.4 Summary Table

Reference	Title	Problem Addressed	Methodology	Results	Limitations
[1] Hu, Zhigang, et al. (2023)	A Set-Associative Cache with Improved Cache Power Efficiency Using Asymmetric Associativity	More cache associativity requires more power	Asymmetrical cache structure with selective enabling	Improved, lower cache power consumption & efficiency	More complex cache controller
[2] M Vinodhini et al. (2024)	Syndrome based multi-bit error detection and correction code using generator and parity-check matrices	Ordinary ECC cannot fix neighbouring multi-bit errors	Syndrome decoder using Generator & parity check matrices	More efficient Multi-bit error correction efficiency	More complex parity and decoder
[3] Shanthi,	Design and Implementation of a Syndrome	SRAM memories have	Encoding, Syndrome	Efficient adjacent multi	More timing and hardware overhead

G., et al. (2025)	based Multi-Bit Error Detection and Correction Code using Generator and Parity-Check Matrices	adjacent fault,burst errors	decoding with matrices	bit error correction	
[4] Zhou, Yulun, et al. (2022)	Implementation of multi-bit error detection and correction codes using generator-parity check matrix technique	Static ECC overhead is redundant	Re-configurable adaptable ECC system and dynamically adaptable modes	Enhanced reliability and timing improvement	More complex mode changing
[5] Joshi, Dinesh, et al. (2026)	ECC enabled cache architecture with modeling of energy and delay	Strong ECC causes higher power and delay overhead	Analytical performance of ECC on time and energy	Well defined ECC overhead trade-offs	No adaptive runtime switching implementation

[6] Saiz-Adalid, Luis-J., et al. (2019)	MCU fault tolerance in SRAMs using low redundancy triple adjacent error correction	Multiple cell upsets destroy neighbors of the affected SRAM cellsbits	Proximity based syndrome decoding	Improved adjacent burst correction with minimum redundancy	More complex decoder for more correction capability
[7] Bender, Michael A., et al.(2023)	Set associative cache and an associativity threshold phenomenon	Increased associativity requires more power and hardware complexity	Comparison of different cache associations with performance	Moderate associativity balance between performance & efficiency	Doesn't mention the ECC integration and how it will improve cache reliability.

Table 2.1 Summary Table

CHAPTER 3: DESIGN AND ARCHITECTURE

3.1 Introduction

Cache memory plays a vital role in modern computing systems to speed up the processor by decreasing the access time of frequently used data and instructions. Processors have extremely high speed compared to the main memory. Thus, accessing main memory for every access will drastically slow down the processor's speed. So cache memory is introduced in between processor and the main memory. With increasing demand for applications like AI, cloud computing, high-performance computing, and embedded system to deal with vast amount of data, robust and energy-efficient cache memories are increasingly essential.

Despite of technology scaling to achieve higher density and processor speed, the scaling has also brought down the reliability of SRAM based cache memories significantly. The transistors in modern silicon technology have become extremely susceptible to radiations and it is a known that soft errors occur in an SRAM cell due to these radiations, or voltage noise and process variations or thermal noise. These errors can cause corruption in one or more bits of data stored within the SRAM cell and subsequently affect the processor execution. A large number of adjacent bits in highly scaled SRAM technology can be upset in close proximity due to a single radiation strike producing a multi bit burst error. Such an error can also occur due to various reasons in addition to radiation strikes, and is generally called multi-bit upset or multi-cell upset.

Traditional ECC schemes such as Parity codes and SECDED codes can only detect/correct single bit error or a subset of multiple bit error condition. They fail in detecting or correcting a large number of adjacent burst errors which become common in a modern cache memory. Modern ECC schemes like PCM codes or MBE-DCC codes exhibit better fault tolerance, but they introduce a lot of

overhead like hardware, area, delay and power consumption. This extra logic will remain on for all the time, causing wastage of power and speed when the severe burst error is not expected during normal operation.

To eliminate these above mentioned drawbacks, the current work suggests a reconfigurable cache memory architecture with adaptive MBE-DCC that can modify its correction capability based on runtime error situations. It uses the two possible correction mechanisms that supports 4-burst and 6-burst modes. The system is in light-weight 4-burst mode in normal condition so that the power consumption is minimized and hardware overhead is limited. In case of excessive adjacent burst errors the controller automatically switches the mode into a strong 6-burst mode to enhance its fault-tolerance and reliability.

The architecture includes the two-way set associative cache organization with tag memory, valid bit logic, cache controller, MBE-DCC encoder, syndrome generator, decoder and adaptive ECC mode manager. The encoder produces the error correctable code (ECC) protected words before storing into cache while the decoder detects the syndrome and provides the error-correction for the bursts during reading operation. It dynamically switches the error correction mechanism depending upon the observed errors during run time using adaptive ECC mode manager module.

The whole system is coded in a synthesizable Verilog HDL and functional simulation and synthesis has been performed. This work has emphasized on increasing the reliability of the cache memory under burst error conditions while reducing power consumption, area and optimizing timing performance, keeping reasonable hardware efficiency. The methodology of designing the whole system, organization of cache, process of integrating ECC in to the system and its functioning is presented in detail in the following chapter.

3.2 Overall Proposed Architecture

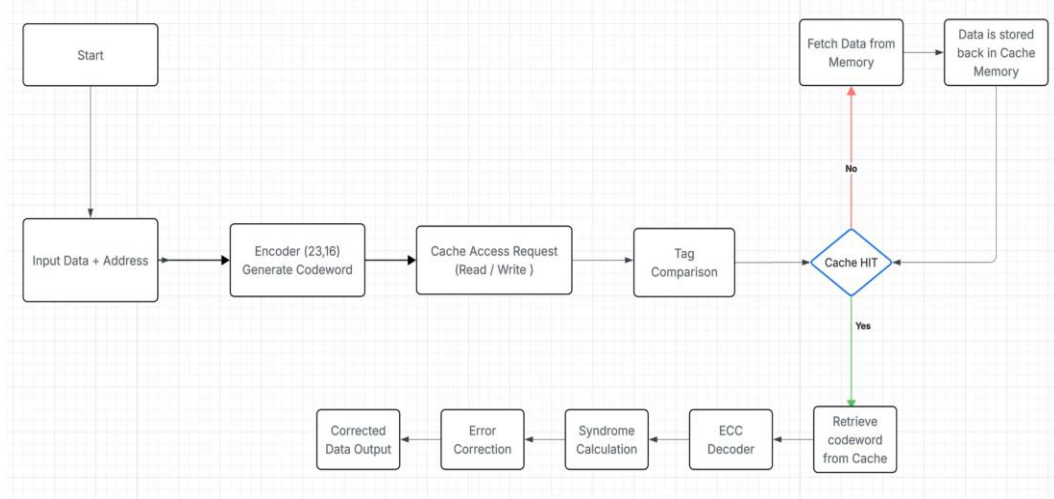


Figure 3.1 Overall Operational Flow of Reconfigurable Cache Memory with ECC Integration

Figure 3.1 illustrates the whole mechanism of proposed architecture of Reconfigurable Cache Memory with an adaptive Multi-Bit Error Detection and Correction Code. The architecture has integrated cache memory and burst error correction codes together so that memory can be made reliable. The whole operation is initiated when a read/write request with data and address is issued by processor.

This processor request comes to cache controller, it decides whether the request is cache write operation or cache read operation. In cache write operation input processor data goes to MBE-DCC encoder and is stored in cache memory. The MBE-DCC encoder generates the parity bits using PCM style generator matrix and concatenates them with input data bits to form an ECC protected word.

The protected word now consists of information bits and parity bits which are necessary for burst error detection and correction. This protected word is then written to the cache memory in the respective address line (within the cache line along with tag and valid bits).

It is a 2-way set-associative cache. When an address from processor arrives in the cache, it is divided into tag, index and offset bits. Based on the index bits the appropriate set in the cache memory is selected. Then tags of the corresponding cached lines of each of the cache ways is compared with the tag bit of the requested address. This is called a cache hit if the requested data is present in the cache. The ECC protected codeword is then directly read out from the respective cache lines.

The required data is read out from main memory by the controller if there is a cache miss by the processor. The controller then feeds the read codeword to an ECC encoder, which adds the parity bits before writing back the data along with the new tag bit & valid bit.

During a read access, when the fetched codeword is input to the syndrome generator & decoder block, syndrome generator produces a syndrome based on the operations done using parity-check matrix. If the value of syndrome bits is zero the data read from the cache is assumed correct, so the same is fed to the processor. However, if the syndrome bit value is non-zero it means that the received codeword has a burst error. The decoder block then locates the position of the errors in the codeword using the syndrome-based error location logic and flips the error bits to get a corrected codeword which is then fed to the processor.

This design has both 4-burst and 6-burst correction mode, it has an adaptive ECC mode manager block. The severity of the bursts can be measured and mode chosen, but since there are usually fewer errors on the cache lines the design runs on 4-burst mode that needs fewer hardware & power.

It is 2-way set-associative cache. When address from the processor come to the cache memory it is divided in to the tag, index and offset bits. Using index bits the location is identified in to the cache memory. Now all the way's tag bits is compared with the tag bits of asked address. If the asked data is found it is called as cache hit. And ECC protected data of that line is given to the processor.

If for the asked address of processor, miss has occurred. Read codeword of asked address is given by the controller of main memory controller. Provided read codeword by the controller is feed to the ECC encoder, parity bits is added and the data is write back to memory along with new tag and valid bit.

During read operation read codeword is sent to syndrome generator & decoder block where using the parity check matrix operation syndrome bit is calculated. If the value of syndrome bits is zero it is assume that, the data from the cache read is correctly taken and it is feed to processor. If syndrome bit values are non-zero then burst error has taken place in read codeword. Using the syndrome-based error location logic position of error is detected and the erroneous bits is flipped and error free data codeword is fed to processor.

Support of 4-burst and 6-burst correction is there, the ECC mode manager block is designed. How bad the burst are identified and the appropriate mode is selected. Number of errors in the cache line will mostly be less so it is design with 4-burst mode for less power.

3.3 Cache Memory Organization

3.3.1 Two-Way Set Associative Cache

The architecture described here utilizes a 2-way set associative cache memory in order to attain an optimal compromise between the hardware costs, the cache hit rate and the speed of access to the memory. The data is grouped into sets within this type of cache memory organization. Each set is further comprised of two cache lines or ways. Each cache line stores the codeword (protected by ECC) and the tag and valid bits corresponding to the stored codeword.

The processor's address is then divided into three bits-the tag bits, the index bits and the offset bits. These bits are used to specify (1) the set in which to find the data at hand; (2) the two tags in the set to be simultaneously compared with the

tags stored in the two cache ways; and (3) the location of the data at hand within the cache line.

There is a cache hit if either of the two tag comparison succeeds and the associated valid bit is set. The conflict miss rate is thereby reduced as compared with direct mapped caches, while the hardware cost penalty is relatively low as compared with fully associative cache designs. Note that the codeword, protected by ECC, is stored at the system architecture level instead of the raw data bits-which is essential for burst error detection and correction during cache read operations.

3.3.2 Tag Memory and Valid Bit Logic

The tag memory is to store the tag bits for every cache line in the cache. As a block of main memory can be mapped to any one of two cache lines in the set of the cache architecture; the reason for two tag bits is to identify whether required data is present in the required cache line.

Every cache line stores two items in addition to data, a valid bit and the value of the tag stored in it. A valid bit tells whether data present in that cache line is valid or not. At the time of resetting all valid bits are set to 0 value. Any valid bit can only be set to 1 only at the time when data is loaded into the cache.

When a cache access is occurred incoming tag bits are compared with both the cache lines of the set. When the tag matches and the valid bit is 1, it is known as cache hit otherwise it is a cache miss and thus the data has to be fetched from the main memory. It is a very critical process of matching tags as an invalid tag matching process will cause to fetch garbage data and will result in the unstable behaviour of the processor.

3.3.3 Cache Controller

Cache Controller: This is the central controller unit in proposed architecture. The Cache controller acts as an interface to the processor, cache memory and ECC units. It controls the cache read/write operations, hit/miss check and tag compare. For every access to the cache it receives the processor address and control signals and performs either a cache read operation or cache write operation based on the read/write request made by the processor. On a read operation it receives the ECC encoded codeword for the specified cache line from the cache memory and then passes it on to the syndrome generator & decoder block. In the case of a write operation it takes in the data from the processor, encodes it using MBE-DCC encoder and then writes the encoded codeword into the cache memory. In the event of a cache replacement it updates the tags and valid bits for the replaced line.

3.4 MBE-DCC Encoder and Decoder Design

3.4.1 MBE-DEC Encoder

The MBE-DCC encoder generates parity bits which is used to detect and correct burst error. The input original data bits are input to the encoder which generate parity bits using PCM-based generator matrix.

The parity generation logic consists usually of a network of combinational XORs that compute the parity value based on desired generator relationships. These parity bits are concatenated with original data bits to form the resulting ECC data codeword.

It is able to perform 4-burst and 6-burst correction depends on operation mode determined by adaptive controller. The processor data block, while being written to the cache, it is encoded each time prior to being written.

Hardware complexity and propagation delay of the encoder is kept as low as possible while satisfying burst correction requirement.

3.4.2 Syndrome Generator and Decoder

For reliability purpose, Syndrome Generator and Decoder is the basic block in the proposed architecture. While reading word from cache, ECC protected word that is read from the cache block is sent to the syndrome generator block. Syndrome generator computes syndrome bits by using parity check matrix operations. If syndrome is zero then the codeword received is free from error. If syndrome is non zero then the decoder detects burst error in the received codeword. Decoder checks pattern of syndromes then error position is determined based on error pattern. Based on error pattern error correction logic flips the bits which are erroneous, thus error corrected word is obtained. Then data bits are extracted and supplied to processor.

3.4.3 4-Burst and 6-Burst Correction Modes

The proposed architecture support 4-burst and 6-burst correction mode. The 4-burst architecture aims to correct medium-level adjacent burst errors with low hardware overhead and low power consumption. As medium-level adjacent burst errors are more commonly found during the normal operations, the system runs in this lightweight mode by default.

The 6-burst architecture supports more powerful error correction for severe-level adjacent burst errors, which are found more frequently inside the SRAM cache memories. It consumes more parity bits and requires complex syndrome decoding logic for stronger error correction capability.

Although more powerful correction mechanism provides better reliability, it consumes more hardware resources, introduces more timing delay and power

consumption. Thus the proposed system could dynamically adjust its correction level according to current error situation at run-time.

3.5 Adaptive ECC Mode Switching Mechanism

3.5.1 Adaptive Mode Manager

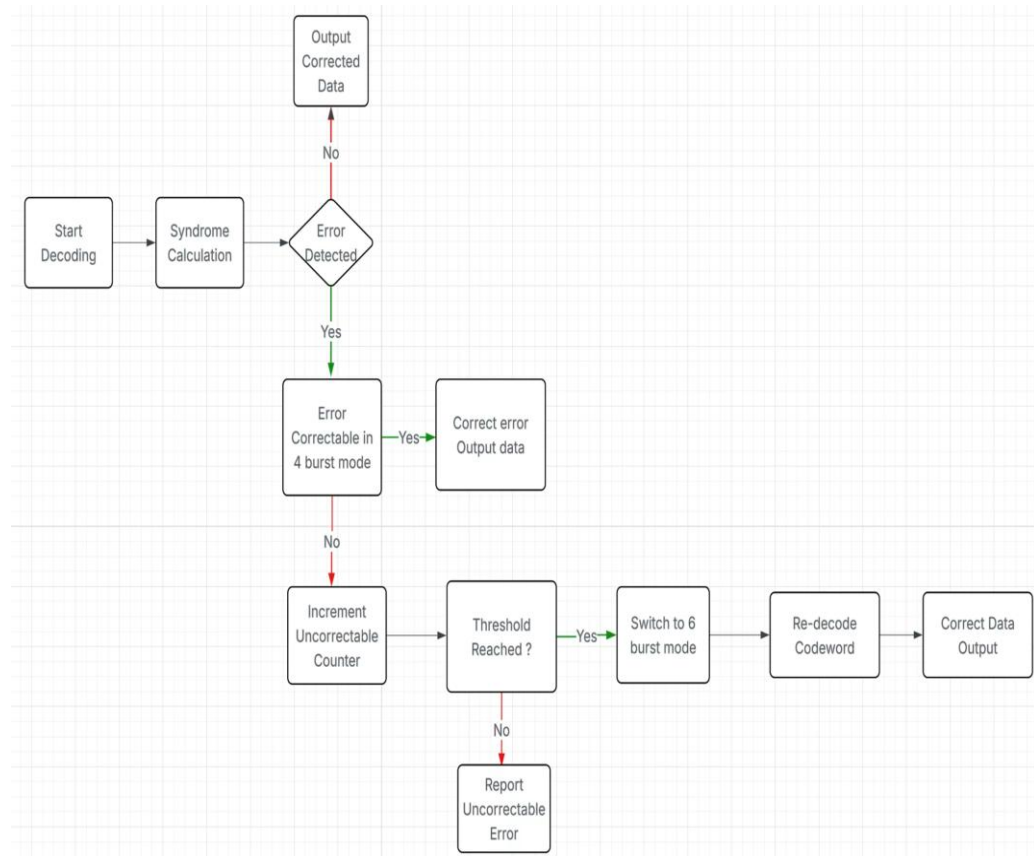


Figure 3.2 Adaptive Burst Error Correction and ECC Mode Switching Flow

The dynamic behavior of the adaptive ECC mode manager for adjusting the error correction capability of proposed cache memory architecture. In order to save the activity and power consumption of hardware, the architecture works on lighter 4-burst correction mode initially. During cache reading operation, the syndrome generator checks if the detected error is within the current mode's

correction capability. If the error can be correctable, the corrected data is directly presented to processor.

In other hand, if the burst error to be corrected is out of the current mode's correction capability, the architecture will perceive it as an uncorrectable error. There is an error counter in the adaptive controller to monitor the severity and frequency of uncorrectable burst error happening inside cache memory.

Once the uncorrectable error number reaches a pre-defined threshold, the adaptive controller automatically switches the architecture to higher 6-burst correction mode. Then, the incorrect codeword can be correctly generated through the higher correction mechanism after changing to a stronger correction mode. Through such dynamic mechanism, the performance, reliability, and power consumption of the architecture can be properly balanced.

3.5.2 Advantages of Adaptive ECC Switching

Compared to the conventional static ECC schemes, the adaptive ECC switching schemes has the following advantages: The design presented is power effective because the unnecessary hardware work can be prevented in the normal condition as light correction logic is active as far as it is necessary.

Another advantage of this architecture is higher reliability against the adjacent burst error compared to static ones, because more powerful correction hardware is active only if faults are dominant. This not only saves average power but also lessens the penalty on normal condition.

Hence, the presented adaptive ECC scheme achieves good trade off among the power, reliability, timing performance, and area for modern SRAM based cache.

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Introduction

The simulation and synthesis of the developed reconfigurable cache system with adaptive MBE-DCC logic has been discussed in this chapter. Functionality of the system was checked and confirmed. The functionality test of developed cache was to check operation of cache fill, cache hit and miss detection, burst error detection and correction capability, syndrome generation and the change over of adaptive ECC modes and cache functionality. The Developed cache system functionality with different burst error pattern application was also tested, to check correctability capability and real time error correction under specified errors and to find the cache performance in real time.

The developed cache system has 4 supported ECC modes which are: 4-Burst Only ECC, 6-Burst Only ECC, adaptive 6B+4B ECC and adaptive 4B+3B ECC. There are two lightly correctable and power & hardware efficient designs implemented, two highly correctable designs implemented to tolerate maximum number of errors for the SRAM based cache memory system, whereas the adaptive ECC schemes to reduce hardware overhead and optimize error correction under burst errors of larger length.

It can be observed from the simulation results that, the designed system performs all functional operations correctly including cache fill, cache hit and miss detection, syndrome bit generation, error detection and correction, uncorrectable error detection, cache refill, ECC mode switching. Beside functional simulation synthesis of the developed system was also performed by Cadence Genus by the use of GPDK045 technology to check the area utilization, power dissipation, performance, gate count and hardware overhead for the adaptive ECC design.

The detailed report of the output waveforms, synthesis reports, compare result and architecture performance is mentioned here in this chapter. The test result

shows that the suggested architecture is suitable to overcome adjacent burst errors in cache and it doesn't result in any hardware over-complexity and time performance degradation.

4.2 Functional Simulation Results of 4-Burst ECC Architecture

4.2.1 Advantages of Adaptive ECC Switching

```
# run 2000ns
===== CACHE FILL =====

[CACHE WRITE] Addr=000001 | Set=1 | Tag=0000
? Writing to WAY0 (empty)

[CACHE WRITE] Addr=000101 | Set=1 | Tag=0001
? Writing to WAY1 (empty)

[CACHE WRITE] Addr=100101 | Set=1 | Tag=1001
? Writing to WAY0 (empty)

===== CACHE OPERATIONS =====
175000 [CACHE] MISS | Addr=000001 -> REFILL
235000 [CACHE] REFILL SUCCESS
266000 [ECC] CORRECTED | Burst=2
Two-bit adjacent error at 22,23
325000 [CACHE] MISS | Addr=000101 -> REFILL
385000 [CACHE] REFILL SUCCESS
416000 [ECC] CORRECTED | Burst=1
Single-bit error at 2
475000 [CACHE] MISS | Addr=100101 -> REFILL
535000 [CACHE] REFILL SUCCESS
566000 [ECC] UNCORRECTABLE ERROR
Error could not be corrected.
```

Figure 4.1 Cache Fill and Initialization in 4-Burst ECC

The waveform shown in figure 4.1 indicates the operation of initial cache filling operation of the proposed 4-burst ECC architecture. In it, processor data is written to the cache memory with parity bits that is calculated from the ECC block. Cache controller identifies an empty cache line and stores the coded data into the cache ways.

The waveform shows the accurate initialization of the cache and successful cache writes. The controller knows the empty locations for storage, and write the

incoming codeword to the right cache line. The cache fill operation also updates the tag and valid bits appropriately so that the cache line could be accessed later.

This simulation demonstrates that the interface between cache controller and ECC encoder is successful when data is written to the cache memory. The input processor data word is transformed to ECC coded codeword and written to the cache memory.

4.2.2 Cache Miss Handling and Burst Error Correction

```

625000 [CACHE] MISS | Addr=000001 -> REFILL
685000 [CACHE] REFILL SUCCESS
716000 [ECC] CORRECTED | Burst=3
Three-bit adjacent error at 18,19,20
775000 [CACHE] MISS | Addr=000101 -> REFILL
835000 [CACHE] REFILL SUCCESS
866000 [ECC] CORRECTED | Burst=4
Four-bit adjacent error at 10,11,12,13
925000 [CACHE] MISS | Addr=100101 -> REFILL
985000 [CACHE] REFILL SUCCESS
1016000 [ECC] CORRECTED | Burst=4
Four-bit adjacent error at 15,16,17,18
1075000 [CACHE] HIT | Addr=000101
1106000 [ECC] CORRECTED | Burst=3
Three-bit adjacent error at 12,13,14

=====
CACHE PERFORMANCE SUMMARY
=====
Total Cache Hits    = 7
Total Cache Misses = 6

=====
ECC RELIABILITY SUMMARY
=====
Uncorrectable Errors = 0

===== SIMULATION COMPLETE =====

```

Figure 4.2 Cache Miss and Error Correction in 4-Burst ECC

This figure depicts the cache miss and burst error correction occurrences in the 4-burst ECC architecture. As there are no hits in the requested cache set, cache misses will be seen. This necessitates the cache controller fetch data from main memory and refill the cache. The figure shows the waveform corresponding to the refill event, ECC correction events, and also clearly depicts that the 4-burst

architecture can correct the burst errors by a 1, 2, 3 and 4 bit adjacent burst error pattern.

There were no uncorrectable faults detected, based upon the syndrome creation and correction occurrences. This tests verify that the 4-burst architecture can correct moderately sized burst errors with only a minimal design. Also these tests verify that it can correct all burst patterns within its detectable error range. For zero uncorrectable error test runs cache performance was determined to be 7 hits and 6 misses (as shown in the ECC reliability summary).

4.3 Functional Simulation Results of 6-Burst ECC Architecture

4.3.1 Cache Operations and Initial ECC Processing

```
# run 2000ns
===== CACHE FILL =====

[CACHE WRITE] Addr=000001 | Set=1 | Tag=0000
? Writing to WAY0 (empty)

[CACHE WRITE] Addr=000110 | Set=2 | Tag=0001
? Writing to WAY0 (empty)

[CACHE WRITE] Addr=100111 | Set=3 | Tag=1001
? Writing to WAY0 (empty)

===== CACHE OPERATIONS =====
175000 [CACHE] HIT | Addr=000001
206000 [ECC] CORRECTED | Burst=1
Single-bit error at 22
265000 [CACHE] MISS | Addr=000101 -> REFILL
325000 [CACHE] REFILL SUCCESS
356000 [ECC] CORRECTED | Burst=2
Two-bit adjacent error at 18,19
415000 [CACHE] MISS | Addr=001001 -> REFILL
475000 [CACHE] REFILL SUCCESS
506000 [ECC] CORRECTED | Burst=2
Two-bit adjacent error at 3,4
565000 [CACHE] MISS | Addr=000001 -> REFILL
625000 [CACHE] REFILL SUCCESS
656000 [ECC] CORRECTED | Burst=5
Five-bit adjacent error at 1,2,3,4,5
715000 [CACHE] MISS | Addr=000101 -> REFILL
775000 [CACHE] REFILL SUCCESS
```

Figure 4.3 Cache Operations and ECC Correction in 6-Burst ECC

The Figure 4.3 shows a description of the way the static 6-burst ECC architecture operates during a cache access. Cache fills are identical to those seen in the 4-burst architecture; with ECC-protected codewords stored in the cache memory. On a cache read, adjacent burst error data are output and presented to the syndrome generator and syndrome information supplied to the higher strength 6-burst decoder. It can be seen from the waveform that single bit adjacent burst error and two bit adjacent burst error have been corrected using syndrome based correction. The 6-burst decoder uses a stronger syndrome based operation than its lightweight correction counterpart due to the larger parity space to allow the stronger bursts to be corrected.

.4.3.2 Higher Order Adjacent Burst Error Correction

```

Two-bit adjacent error at 3,4
    565000 [CACHE] MISS | Addr=000001 -> REFILL
    625000 [CACHE] REFILL SUCCESS
    656000 [ECC] CORRECTED | Burst=5
Five-bit adjacent error at 1,2,3,4,5
    715000 [CACHE] MISS | Addr=000101 -> REFILL
    775000 [CACHE] REFILL SUCCESS
    806000 [ECC] CORRECTED | Burst=6
Six-bit adjacent error at 7,8,9,10,11,12
    865000 [CACHE] MISS | Addr=001001 -> REFILL
    925000 [CACHE] REFILL SUCCESS
    956000 [ECC] CORRECTED | Burst=4
Four-bit adjacent error at 10,11,12,13
    1015000 [CACHE] HIT | Addr=000101
    1046000 [ECC] CORRECTED | Burst=4
Four-bit adjacent error at 6,7,8,9

=====
CACHE PERFORMANCE SUMMARY
=====
Total Cache Hits    = 7
Total Cache Misses = 5

=====
ECC RELIABILITY SUMMARY
=====
Uncorrectable Errors = 0

===== SIMULATION COMPLETE =====

```

Figure 4.4 Higher Order Burst Error Correction in 6-Burst ECC

In above figure 4.4 is depicted the 6 larger adjacent burst error correction with the stronger 6-burst decoder. The suggested 6-burst structure does in fact manage to correct a 4-bit, 5-bit and 6-bit adjacent burst error corruption into cache memory system contents when written into it. The displayed waveform depicts a correctability of the corrupted codeword even under heavy burst fault condition. As stated before the stronger decoder is able to locate the location of the bit errors inside corrupted codeword, to be then corrected via syndrome-based error location. Controller does refresh the cache via cache refill on cache miss.

Hardware overhead and correction complexity of the stronger 6-burst decoder is slightly higher than the light weight ECC ones but reliability for a severe multi cell upset inside SRAM cache is greater. The last simulated results reveals 7 hits and 5 misses in cache memory, interestingly there is no uncorrectable errors as can be seen on the ECC reliability summary.

4.4 Functional Simulation Results of Adaptive 6B+4B ECC Architecture

4.4.1 Initial Cache Operations in Lightweight 4-Burst Mode

```

===== CACHE FILL =====

[CACHE WRITE] Addr=000001 | Set=2 | Tag=0000
? Writing to WAY0 (empty)

[CACHE WRITE] Addr=000110 | Set=3 | Tag=0001
? Writing to WAY0 (empty)

[CACHE WRITE] Addr=100111 | Set=4 | Tag=1001
? Writing to WAY0 (empty)

[CACHE READ] Addr=000110 | Set=3 | Tag=0001
? HIT in WAY0
      86000 [CACHE] HIT | Addr=000110
[INFO] CACHE HIT

===== ECC PROCESSING (DATA2) =====
Single-bit error at 23
      146000 [ECC] CORRECTION | Burst=1

[CACHE READ] Addr=100111 | Set=4 | Tag=1001
? HIT in WAY0
      186000 [CACHE] HIT | Addr=100111
[INFO] CACHE HIT

===== ECC PROCESSING (DATA3) =====
Two-bit adjacent error at 20,21
      246000 [ECC] CORRECTION | Burst=2

[CACHE READ] Addr=000001 | Set=2 | Tag=0000
? HIT in WAY0

```

Figure 4.5 Initial Cache Operations in Adaptive 6B+4B ECC

Figure 4.5 depicts basic operation of the adaptive 6B+4B ECC. In normal mode of operation of adaptive system it operate on light 4-burst correction mode so to save hardware resources and decrease power.

Cache controller fill up the cache correctly and reads back the ECC protected codeword when read data from the cache. Waveform show successful correction of 1-bit and two-bit adjacent burst error on the lighter hardware of correction. In no case heavier hardware of correction is enabled.

4.4.2 Detection of Uncorrectable Errors

```

[CACHE READ] Addr=000001 | Set=2 | Tag=0000
? HIT in WAY0
      286000 [CACHE] HIT | Addr=000001
[INFO] CACHE HIT

===== ECC PROCESSING (DATA1) =====
Error could not be corrected.
      346000 [ECC] UNCORRECTABLE ERROR

[CACHE READ] Addr=000110 | Set=3 | Tag=0001
? HIT in WAY0
      386000 [CACHE] HIT | Addr=000110
[INFO] CACHE HIT

===== ECC PROCESSING (DATA2) =====
Error could not be corrected.
      446000 [ECC] UNCORRECTABLE ERROR

[CACHE READ] Addr=100111 | Set=4 | Tag=1001
? HIT in WAY0
      486000 [CACHE] HIT | Addr=100111
[INFO] CACHE HIT

===== ECC PROCESSING (DATA3) =====
Error could not be corrected.
      546000 [ECC] UNCORRECTABLE ERROR
Switching to 6 burst ECC

```

Figure 4.6 Detection of Uncorrectable Errors in Adaptive ECC

The Figure 4.6 illustrates the appearance of adjacent serious burst error which exceeds the correction capability of the lightweight decoder which is turned on. The architecture recognizes these errors as an uncorrectable error.

The adaptive mode manager constantly monitors occurrence of these serious errors using an internal error counter. When an uncorrectable error keeps happening over and over again, the adaptive controller is ready to transition to a more powerful correction mode.

The waveform is the confirmation that the serious burst error can be correctly detected and the adaptive error monitoring mechanism which is built within the proposed architecture is working as expected.

4.4.3 Adaptive Switching to Stronger 6-Burst Mode

```
[CACHE READ] Addr=000001 | Set=2 | Tag=0000
? HIT in WAY0
      586000 [CACHE] HIT | Addr=000001
[INFO] CACHE HIT

===== ECC PROCESSING (DATA1) =====
Five-bit adjacent error at 8,9,10,11,12
      646000 [ECC] CORRECTION | Burst=5

[CACHE READ] Addr=000110 | Set=3 | Tag=0001
? HIT in WAY0
      686000 [CACHE] HIT | Addr=000110
[INFO] CACHE HIT

===== ECC PROCESSING (DATA2) =====
Single-bit error at 23
      746000 [ECC] CORRECTION | Burst=1

[CACHE READ] Addr=100111 | Set=4 | Tag=1001
? HIT in WAY0
      786000 [CACHE] HIT | Addr=100111
[INFO] CACHE HIT

===== ECC PROCESSING (DATA3) =====
Two-bit adjacent error at 22,23
      846000 [ECC] CORRECTION | Burst=2

[CACHE READ] Addr=100111 | Set=4 | Tag=1001
? HIT in WAY0
      1186000 [CACHE] HIT | Addr=100111
[INFO] CACHE HIT

===== ECC PROCESSING (DATA3) =====
Error could not be corrected.
      1246000 [ECC] UNCORRECTABLE ERROR

[CACHE READ] Addr=000001 | Set=2 | Tag=0000
? HIT in WAY0
      1286000 [CACHE] HIT | Addr=000001
[INFO] CACHE HIT

===== ECC PROCESSING (DATA1) =====
Error could not be corrected.
      1346000 [ECC] UNCORRECTABLE ERROR
Switching to 6 burst ECC

[CACHE READ] Addr=100111 | Set=4 | Tag=1001
? HIT in WAY0
      1386000 [CACHE] HIT | Addr=100111
[INFO] CACHE HIT

===== ECC PROCESSING (DATA3) =====
Six-bit adjacent error at 1,2,3,4,5,6
      1446000 [ECC] CORRECTION | Burst=6

[CACHE READ] Addr=000110 | Set=3 | Tag=0001
? HIT in WAY0
      1486000 [CACHE] HIT | Addr=000110
[INFO] CACHE HIT

===== ECC PROCESSING (DATA2) =====
Five-bit adjacent error at 19,20,21,22,23
      1546000 [ECC] CORRECTION | Burst=5
```

Figure 4.7 Adaptive Switching to 6-Burst ECC Mode

Figure 4.7 shows adaptive switching operation of the proposed architecture. The stronger 6-burst correction mode is enabled by adaptive controller when number of uncorrectable error is above a threshold value. After switching the architecture can now correct five-bit and six-bit adjacent burst errors that were not able to be corrected using lightweight correction logic before. With this kind of adaptive mechanism the runtime performance will be better because the stronger correction logic will be activated at severe fault conditions occur time after time.

4.4.4 Runtime Return to Lightweight ECC Mode

```
[CACHE READ] Addr=000110 | Set=3 | Tag=0001
? HIT in WAY0
      886000 [CACHE] HIT | Addr=000110
[INFO] CACHE HIT

===== ECC PROCESSING (DATA2) =====
Two-bit adjacent error at 20,21
      946000 [ECC] CORRECTION | Burst=2
Switching back to 4 burst ECC

[CACHE READ] Addr=000001 | Set=2 | Tag=0000
? HIT in WAY0
      986000 [CACHE] HIT | Addr=000001
[INFO] CACHE HIT

===== ECC PROCESSING (DATA1) =====
Single-bit error at 18
      1046000 [ECC] CORRECTION | Burst=1

[CACHE READ] Addr=000110 | Set=3 | Tag=0001
? HIT in WAY0
      1086000 [CACHE] HIT | Addr=000110
[INFO] CACHE HIT

===== ECC PROCESSING (DATA2) =====
Error could not be corrected.
      1146000 [ECC] UNCORRECTABLE ERROR

[CACHE READ] Addr=000001 | Set=2 | Tag=0000
? HIT in WAY0
      1586000 [CACHE] HIT | Addr=000001
[INFO] CACHE HIT

===== ECC PROCESSING (DATA1) =====
Single-bit error at 23
      1646000 [ECC] CORRECTION | Burst=1

[CACHE READ] Addr=000110 | Set=3 | Tag=0001
? HIT in WAY0
      1686000 [CACHE] HIT | Addr=000110
[INFO] CACHE HIT

===== ECC PROCESSING (DATA2) =====
Two-bit adjacent error at 20,21
      1746000 [ECC] CORRECTION | Burst=2

[CACHE READ] Addr=100111 | Set=4 | Tag=1001
? HIT in WAY0
      1786000 [CACHE] HIT | Addr=100111
[INFO] CACHE HIT

===== ECC PROCESSING (DATA3) =====
Three-bit adjacent error at 5,6,7
      1846000 [ECC] CORRECTION | Burst=3
Switching back to 4 burst ECC
```

Figure 4.8 Runtime Switching Back to Lightweight Mode

Shows architecture switching dynamically between the light 4-burst correction mode and robust correction mode once severe error conditions pass. Once the more aggressive mode has corrected the burst errors it automatically switches back to the light mode to reduce activity and power. Shows that it is possible to switch the modes whilst the run is in operation as we see an in run switch and that the activity on the cache does not stop when this occurs. The 4 modes single bit adjacent, double bit adjacent and the two burst modes triple and four bit adjacent burst were still correctable by the architecture in the light mode.

In the summary of simulation results we have that there was a twenty hit zero miss architecture, the required data was present in the cache at the time of the read. The summary for ECC reliability states that there were 6 uncorrectable bits detected in the run-time until the 4-burst mode was up and running.

4.5 Functional Simulation Results of Adaptive 4B+3B ECC Architecture

4.5.1 Initial Cache Access and Lightweight ECC Operation

```

===== ECC PROCESSING (DATA2) =====
Two-bit adjacent error at 22,23
          746000 [ECC] CORRECTION | Burst=2
Switching back to 3 burst ECC

[CACHE READ] Addr=100111 | Set=4 | Tag=1001
? HIT in WAY0
          786000 [CACHE] HIT | Addr=100111
[INFO] CACHE HIT

===== ECC PROCESSING (DATA3) =====
Three-bit adjacent error at 21,22,23
          846000 [ECC] CORRECTION | Burst=3

=====
CACHE PERFORMANCE SUMMARY
=====
Total Cache Hits    = 8
Total Cache Misses = 0

=====
ECC RELIABILITY SUMMARY
=====
Uncorrectable Errors Detected = 3

=====
SIMULATION COMPLETED
=====

```

Figure 4.9 Cache Operations in Adaptive 4B+3B ECC

The Figure 4.9 shows the adaptive 4B+3B ECC structure in fig.4.9. The structure is at the lightweight 3-burst correction mode in order to keep hardware area and power consumption. In the figure it shows that cache is filled successfully, and then burst error is detected and syndrome is computed. A grave burst error that exceeds the lightweight correction capacity is then marked as uncorrectable fault.

4.5.2 Adaptive Switching to 4-Burst Correction Mode

```

===== CACHE FILL =====

[CACHE WRITE] Addr=000001 | Set=2 | Tag=0000
? Writing to WAY0 (empty)

[CACHE WRITE] Addr=000110 | Set=3 | Tag=0001
? Writing to WAY0 (empty)

[CACHE WRITE] Addr=100111 | Set=4 | Tag=1001
? Writing to WAY0 (empty)

[CACHE READ] Addr=000110 | Set=3 | Tag=0001
? HIT in WAY0
      86000 [CACHE] HIT | Addr=000110
[INFO] CACHE HIT

===== ECC PROCESSING (DATA2) =====
Error could not be corrected.
      146000 [ECC] UNCORRECTABLE ERROR

[CACHE READ] Addr=100111 | Set=4 | Tag=1001
? HIT in WAY0
      186000 [CACHE] HIT | Addr=100111
[INFO] CACHE HIT

===== ECC PROCESSING (DATA3) =====
Error could not be corrected.
      246000 [ECC] UNCORRECTABLE ERROR

```

Figure 4.10 Adaptive Switching to 4-Burst Mode

Figure 4.10. Switchover from 3-burst correction mode to 4-burst correction mode in lightweight correction. The adaptive controller correctly detected the repeating uncorrectable errors and activated the higher decoding capability hardware. The waveform indicates the correction of the 4 consecutive bits burst error. It demonstrated the feasibility of adaptive controller to improve the reliability in accordance to runtime error condition.

4.5.3 Runtime Return to Lightweight 3-Burst Mode

```

[CACHE READ] Addr=000001 | Set=2 | Tag=0000
? HIT in WAY0
      286000 [CACHE] HIT | Addr=000001
[INFO] CACHE HIT

===== ECC PROCESSING (DATA1) =====
Error could not be corrected.
      346000 [ECC] UNCORRECTABLE ERROR
Switching to 4 burst ECC

[CACHE READ] Addr=000110 | Set=3 | Tag=0001
? HIT in WAY0
      386000 [CACHE] HIT | Addr=000110
[INFO] CACHE HIT

===== ECC PROCESSING (DATA2) =====
Four-bit adjacent error at 3,4,5,6
      446000 [ECC] CORRECTION | Burst=4

[CACHE READ] Addr=100111 | Set=4 | Tag=1001
? HIT in WAY0
      486000 [CACHE] HIT | Addr=100111
[INFO] CACHE HIT

===== ECC PROCESSING (DATA3) =====
Single-bit error at 23
      546000 [ECC] CORRECTION | Burst=1

```

Figure 4.11 Runtime Return to 3-Burst ECC Mode

Figure 4.11 depicts the switching over from lightweight 3-burst correction to adaptive mode during heavy error conditions. The hardware is able to perform correct moderate burst errors without activating the high-overhead correction logic. The activity on the bus is limited and also there is significant saving on hardware.

The above figure supports the claims for an efficient runtime adaptability of the proposed adaptive ECC controller.

4.5.4 Final Cache and Reliability Summary

```

[CACHE READ] Addr=000001 | Set=2 | Tag=0000
? HIT in WAY0
          586000 [CACHE] HIT | Addr=000001
[INFO] CACHE HIT

===== ECC PROCESSING (DATA1) =====
Single-bit error at 23
          646000 [ECC] CORRECTION | Burst=1

[CACHE READ] Addr=000110 | Set=3 | Tag=0001
? HIT in WAY0
          686000 [CACHE] HIT | Addr=000110
[INFO] CACHE HIT

===== ECC PROCESSING (DATA2) =====
Two-bit adjacent error at 22,23
          746000 [ECC] CORRECTION | Burst=2
Switching back to 3 burst ECC

[CACHE READ] Addr=100111 | Set=4 | Tag=1001
? HIT in WAY0
          786000 [CACHE] HIT | Addr=100111
[INFO] CACHE HIT

===== ECC PROCESSING (DATA3) =====
Three-bit adjacent error at 21,22,23
          846000 [ECC] CORRECTION | Burst=3

=====

```

Figure 4.12 Performance Summary of Adaptive 4B+3B ECC

The last simulation summary report shows 8 hits and 0 misses on adaptive state and the ECC reliability summary report shows 3 uncorrectable errors are occurred before the better correction state is turn on successful.

CHAPTER 4: RESULTS AND DISCUSSION

MAY-2026

Showed the adaptive 4B+3B architecture is lower complexity and power consuming compare with high strength ECC architecture.

4.6 Synthesis Results and Comparative Analysis



Figure 4.13 Overall Synthesis Summary of ECC Configurations

All ECC configurations have been synthesized for hardware efficiency using Cadence Genus with standard-cell library GPDK045 and its timing performance, area utilization, power and hardware overheads are presented in the synthesis reports for the 4-Burst Only, 6-Burst Only, Adaptive 6B+4B and Adaptive 4B+3B configurations.

In order to maintain lower power and minimum area, 4-Burst Only has a lower static power consumption and lesser hardware due to the simpler correction and very lightweight syndrome decoding logic and achieved the smallest hardware overhead of 4606.3 m area and 651 W total power. With these specifications it has the shortest critical path delay and thus the highest clock frequency of 200.6 MHz and a critical path delay of 4986 ps.

As expected, the 6-Burst Only configuration has a relatively greater hardware overhead than the 4-Burst Only configuration due to strong syndrome decoding logic which would take higher power. Nevertheless, the area and power consumption are negligibly greater (4758 m area and 683.7 W power) while having a significantly better correction mechanism for longer bursts.

As estimated, both adaptive ECC architectures had high static overheads due to the multiple decoders used, an adaptive controller and mode management circuitry and also an output selection multiplexer. Adaptive 6B+4B consumes the most static power of 895.3 W and utilizes a large area of 6265.9 m while Adaptive 4B+3B consumed a power of 855.1 W and utilizes an area of 5995.2 m respectively. However since adaptive ECC's only one decoder will ever be activated at a time in actual run time it would have much less runtime power consumption compared to the estimates of static synthesis power.

4.7 Timing Analysis

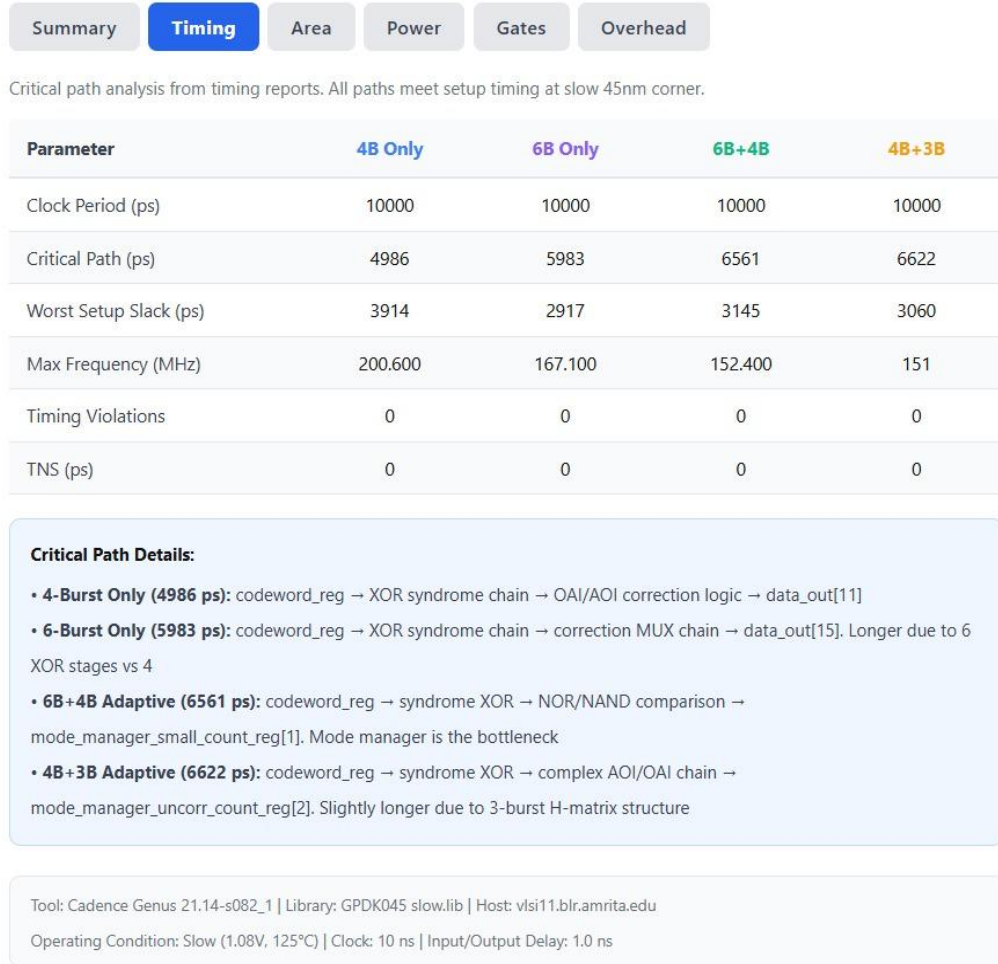


Figure 4.14 Timing Comparison of ECC Architectures

It can be observed that all of the implemented ECC designs satisfy timing constraints. Among all of them, 4-Burst Only configuration was with minimal critical path due to least complexity in syndromes computation & correction circuit. A slight delay found in 6-Burst Only was due to more length of syndrome computation chain and more powerful correction. Both the adaptive versions got a slight delay because of routing through mode switching and adaptive controller circuitry. All the designs were observed to be with setup slack and no violations although at a hardware cost.

4.8 Area and Power Analysis

Summary	Timing	Area	Power	Gates	Overhead
Physical library area in μm^2 using GPDK045 45nm standard cells.					
Parameter	4B Only	6B Only	6B+4B	4B+3B	
Total Cell Count	762	813	1294	1198	
Sequential Instances	276	276	285	285	
Combinational Instances	486	537	1009	913	
Cell Area (μm^2)	3641.488	3729.459	4663.025	4507.338	
Net Area (μm^2)	964.848	1028.574	1602.913	1487.865	
Total Area (μm^2)	4606.335	4758.033	6265.937	5995.203	
Sequential Area %	71.100	69.400	57.300	59.200	
Logic Area %	28.200	29.700	41.100	39.300	
Inverter Area %	0.700	0.900	1.700	1.500	

Area Overhead Analysis:

- Going from 4-Burst to 6-Burst only adds $88 \mu\text{m}^2$ cell area (+2.4%) — confirming 6-burst decoder is only marginally larger than 4-burst
- 6B+4B Adaptive vs 6-Burst Only: $+934 \mu\text{m}^2$ cell area (+25%) — cost of adding second decoder + mode manager + MUX selector
- Sequential logic dominates in single-mode configs (71% for 4-burst, 69% for 6-burst) because cache storage FFs are the largest component
- Logic area grows significantly in adaptive configs (41%, 39%) — dual decoders add large combinational syndrome computation blocks
- Net area increase in adaptive configs is disproportionately large — more logic means more routing complexity and longer interconnects

Figure 4.15 Area Comparison of ECC Architectures

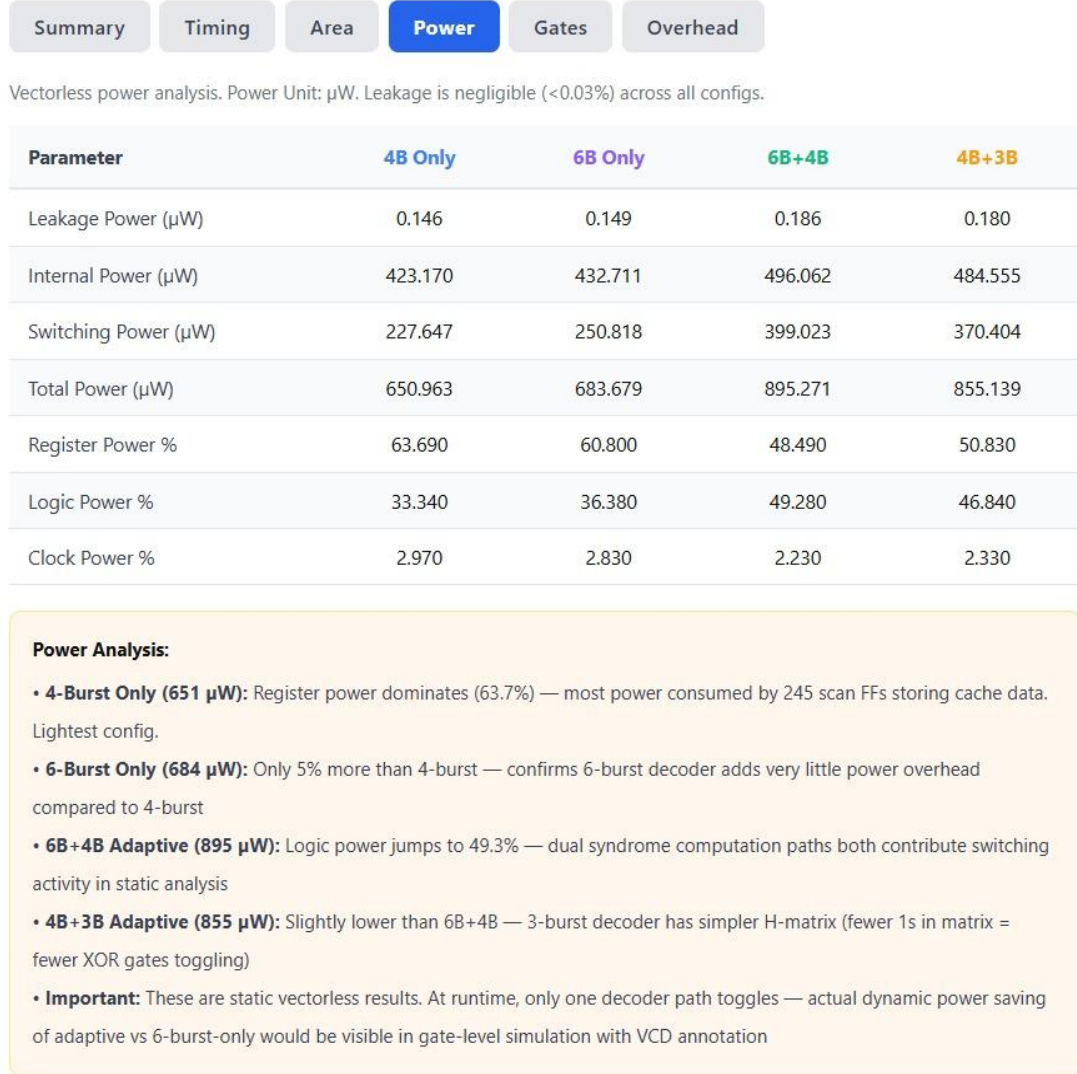


Figure 4.16 Power Comparison of ECC Architectures

The area analysis has proven that lightweight ECCs use less hardware as compared to the more powerful/adaptive ECCs, this is because the calculation of syndrome for lightweight ECC is not complicated and also the decoding logic is less complicated than for adaptive ECC. For the more powerful and adaptive ECC architectures, multiplexers, more XOR networks, syndrome comparing

logic and the adaptive mode control circuitries are synthesized to achieve more robust and adaptive architectures.

The power analysis have proven that register power is the maximum as the flip-flops used for storing the cache consumes maximum switching activity. It is found that the adaptive ECC consumes more power compared to the non adaptive ECC, as multiple syndrome path switching control logic are synthesized in the adaptive ECC.

The performance analysis has shown that runtime is better in adaptive ECC architecture because the strong correction circuit only comes into action when catastrophic fault condition happens and thus the power consumption is controlled to meet better power and reliability trade-off for the adaptive ECC architectures designed.

4.9 Gate-Level and Overhead Analysis

Summary	Timing	Area	Power	Gates	Overhead
Gate-level cell usage extracted from gates reports. Reflects ECC complexity directly.					
Parameter	4B Only	6B Only	6B+4B	4B+3B	
XOR2X1	54	46	62	54	
XOR2X2	17	17	19	15	
XNOR2X1	23	25	44	61	
Total XOR/XNOR	94	89	128	130	
NAND (all types)	79	83	151	129	
NOR (all types)	30	32	95	88	
MUX (MX2+MX12+MX3)	7	10	26	24	
INV gates	38	47	112	97	
SDFF (Scan Flip-Flops)	245	245	247	247	
DFF (Regular Flip-Flops)	31	31	38	38	
AOI gates (all types)	55	60	116	108	
OAI gates (all types)	62	58	122	74	

Figure 4.17 Gate-Level Complexity Analysis

Gate-Level Observations:

- **XOR/XNOR count directly reflects syndrome computation complexity:** $4B=94$, $6B=89$, $6B+4B=128$, $4B+3B=130$.

Adaptive configs nearly double XOR count

- **6-Burst has fewer XOR gates than 4-Burst (89 vs 94)** — Genus found a more efficient factored form for the 6-burst syndrome computation during synthesis optimisation

- **4B+3B has highest XNOR2X1 count (61)** — directly reflects the Li et al. (23,16) 3-burst H-matrix structure which has more XNOR-friendly column patterns

- **Scan FFs (Sdff) dominate sequential area** in all configs — 245-247 scan flip-flops used for cache data storage across all configurations

- **MUX count doubles in adaptive configs (7→26/24)** — reflects decoder output selection MUX and mode_reg pipeline MUX added for adaptive switching

- **AOI/OAI gate counts grow significantly** in adaptive configs — these implement the syndrome comparison and mode switching decision logic

Tool: Cadence Genus 21.14-s082_1 | Library: GPDK045 slow.lib | Host: vlsi11.blr.amrita.edu

Operating Condition: Slow (1.08V, 125°C) | Clock: 10 ns | Input/Output Delay: 1.0 ns

Figure 4.18 Overview of Gate-Level Observations

From gate level view, we can also note the component used the most for XOR/XNOR gate is calculation of syndrome, because many ECC are based on syndrome calculation and parity generation and the adaptive scheme use more XOR/XNOR gate in the networks, normally the adaptive scheme needs more path to decode.



Overhead comparison showing cost of upgrading from one configuration to another.

Parameter	4B Only	6B Only	6B+4B	4B+3B
Total Area vs 4B Only (μm^2)	—	+151.7	+1659.6	+1388.9
Total Area vs 4B Only (%)	—	+3.3%	+36.0%	+30.2%
Total Area vs 6B Only (μm^2)	-151.7	—	+1507.9	+1237.2
Total Area vs 6B Only (%)	-3.2%	—	+31.7%	+26.0%
Total Power vs 4B Only (μW)	—	+32.7	+244.3	+204.2
Total Power vs 4B Only (%)	—	+5.0%	+37.5%	+31.4%
Total Power vs 6B Only (μW)	-32.7	—	+211.6	+171.5
Total Power vs 6B Only (%)	-4.8%	—	+30.9%	+25.1%
Critical Path vs 4B Only (ps)	—	+997	+1575	+1636
Critical Path vs 6B Only (ps)	-997	—	+578	+639

Figure 4.19 Hardware Overhead Comparison

Overhead Summary — What Each Upgrade Costs:

- **4-Burst → 6-Burst Only:** +151.7 μm^2 area (+3.3%), +32.7 μW power (+5.0%), +997 ps critical path (+20%). Very low cost for stronger single-mode protection.
- **6-Burst Only → 6B+4B Adaptive:** +1507.9 μm^2 area (+31.7%), +211.6 μW power (+30.9%), +578 ps critical path (+9.7%). High static cost but enables runtime efficiency.
- **6-Burst Only → 4B+3B Adaptive:** +1237.2 μm^2 area (+26.0%), +171.5 μW power (+25.1%), +639 ps critical path (+10.7%). Slightly cheaper than 6B+4B adaptive.
- **6B+4B → 4B+3B Adaptive:** -270.7 μm^2 area (-4.3%), -40.1 μW power (-4.5%), +61 ps critical path (+0.9%). Marginal difference — 3-burst slightly simpler than 6-burst.
- **4-Burst Only → 6B+4B Adaptive:** +1659.6 μm^2 area (+36.0%), +244.3 μW power (+37.5%). Full cost of adding adaptive capability over simplest baseline.

Figure 4.20 Overview of Overhead Summary

The overhead calculation shows that there is only minimal hardware overhead required to adapt a lightweight complex 4-burst error correction circuit to a stronger complex 6-burst error correction circuit. The area and power overhead is relatively high compared to fixed complexity designs, mainly due to the two decoders placed on a single chip, and also due to run time switching circuit to toggle between the two decoders. However this is outweighed by the fact that they offer adaptability during run time and more robust fault tolerant operation. Thus the advanced SRAM based cache memories which are being developed now with various levels of reliability will tremendously benefit by adaptive architectures.

4.10 Overall Discussion

Based on simulation and synthesis results, we found that the designed reconfigurable cache memory system based on Adaptive MBE-DCC is working correctly. The functional simulation was able to cover every aspect including cache operations, syndrome generation, burst error detection, burst error correction, cache refill, and Adaptive ECC mode selection under different run-

time fault injections conditions. From Static 4-burst architecture, it possesses lowest area, power and speed due to simple decoder logic and less complex syndromes, but it has limited correction ability only to moderate adjacent burst errors.

From Static 6-burst Architecture, it provides higher fault tolerance, it can correct the worst possible adjacent burst errors, no uncorrectable faults will happen; nevertheless, it works with a high strength correction hardware all the time during normal working state. So it leads to high switching power, high switching activity and high timing overhead. Most importantly, we discover that Adaptive 6B+4B ECC Architecture results in the best trade-off of actual system reliability and hardware cost. The Cadence Genus reports give a static hardware analysis for the entire circuit because it synthesized all possible switching logic, multiplexers, and adaptive controller to form a complex circuit. Therefore, Adaptive architectures consumed more area and power than static ECC architectures. At runtime, only one path will be in operation. During normal operation state, Adaptive 6B+4B Architecture works at 4-burst lightweight mode. It will reduce switching activity to save dynamic power. Only when the consecutive severe adjacent bursts exceeding the capability of the lightweight 4-burst mode, the 6B strong decoder will be enabled to achieve high correction power.

The major benefit of the proposed architecture is its ability to adapt itself at runtime. Since this adaptive architecture utilizes stronger correction hardware such as static 6-burst ECC only when necessary and switches to the expensive ones, the practical runtime performance of the Adaptive 6B+4B architecture is much better than the performance result calculated from the static synthesis only.

The adaptive behaviour of the architecture is clearly showed by the simulation results. Initially, this architecture utilized the low-weight 4-burst correction logic to deal with medium range burst error. When it was found that it was constantly

receiving strong burst errors, the adaptive controller automatically enabled the higher-weight 6-burst logic to fix five-bit and six-bit consecutive burst errors. And after the strong burst error condition ended, it again switched back to the low-weight architecture to save hardware power.

Compared to static 6-burst ECC, the Adaptive 6B+4B architecture better in terms of :

Dynamic run time improvements in reliability

Lower average switching activity

Greater power efficiency in practice

Lower waste of the decoder power

Smart utilization of hardware resources

Greater ability to cope with the non-static fault conditions

Similar to the Adaptive 4B+3B architecture, the Adaptive 4B+3B architecture had reasonable runtime switching performance, at lower hardware complexity and lower power consumption. However, the maximal correction performance of the adaptive 4B+3B architecture is still less than the adaptive 6B+4B architecture, as the four-bit adjacent burst correction capability exists only when switched.

Timing Analysis results further confirm that all proposed designs meet the timing requirements without any failure. Although adaptive structures cause additional hardware overhead (e.g., multiple decoder paths and switching logic), this hardware overhead seems reasonable because a considerable improvement on runtime performance, which is extremely important to be in real world system. To conclude, among all the above designs, the Adaptive 6B+4B ECC architecture shows the best performance for current SRAM-based cache memory applications, and it is more suitable to processor cache than those conventional static ECC architectures.

CHAPTER 5: CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

In this project, a Reconfigurable Cache Memory incorporating Adaptive MBE-DCC ECC has been designed, implemented and synthesized. The designed system proves that adaptive burst-error correction can be applied in a realistic 2-way set-associative cache memory with reasonable hardware costs and with quantifiable operational advantages.

The four synthesized implementations using Cadence Genus 21.14 with the GPDK045 45 nm standard cell library at the slow corner successfully meet timing without any violations for a clock period of 10 ns and hence, prove the fabricating capability at the specified technology. Main contributions and conclusions are:

The reconfigurable cache has been built and the adaptive mode manager implemented and put on-chip to contain the 2 types of decoders (4B and 6B) on the same piece of silicon, and to be able to switch modes with the level of protection using a 3 error hysteresis counter.

The synthesis report demonstrates how much static cost this addition brings. The adaptive mode costs 36% more area and 37.5% more power (comparing config 2: 6B+4B adaptive and config 1: 4-burst). This is the manufacturing cost, not the power consumed by a given state.

The switching power from the synthesis reports show how much dynamic benefit this brings. In normal mode (4B) the 4B decoder uses 227.65W to switch while the 6B decoder uses 250.82W. This means that there are 23.17W less being used every cycle for the entire life time of the system.

The critical path of the 4B decoder is 997ps faster than the 6B decoder. Any read access that is taken when operating in normal mode (4B) is therefore 997ps faster

than that that would be if a system was set to 6B mode for the life time of the system.

Adding the 3B, 4B system instead (Config 3) offers a power and area saving adaptive mode to the 6B, 4B system Config 2 by saving 270.73m (4.3% area) and 40.13W (4.5% power).

Config 3 cannot operate to the same level of escalated protection (3 error) however. All synthesis reports resulted in 0 timing violations and 0 total negative slack; it was seen that 0 QOR were obtained and therefore the system could be reliably added.

5.2 Future Scope

Some potential directions in which to take this work further are suggested:

Gate-Level Simulation with VCD Annotation: The most critical is the subsequent step of performing gate-level simulation of the post-synthesized netlist with back-annotated switching activity (VCD form) using a power analysis tool such as Cadence Joules or Synopsys PrimeTime PX that would provide precise runtime switching power measurements of both normal (4-burst mode) and accelerated (6-burst mode) operations, and not just estimations, accurately calculating the adaptive gain.

Physical Design and Place-and-Route: Physical design up to logical synthesis level has been carried out. The final sign-off results on timing, area, and power using full physical design using Cadence Innovus with floorplanning, placement, clock tree synthesis and routing will be accurate post-route timing, area, power measurements with parasitic-aware values.

Extended Cache Organization: the cache used here is of a 2-way 64-set organization with 16-bit words. An increase in the associativity, number of sets

as well as 64-bit words (and therefore codewords) is a better representation of LLC organization of modern CPUs.

Extended ECC Mode Range: Providing at least 3 correction strength options (e.g., 2-burst, 4-burst, 8-burst) with better tracking of error history at the mode manager will allow a better balance between protection and overhead.

FPGA Prototype: An FPGA implementation of the adaptive cache implemented on a Xilinx Virtex or Intel Arria FPGA, subjected to neutron or alpha-particle irradiation, could demonstrate the functional operation of the adaptive switching mechanism in a true radiation environment via hardware-in-the-loop simulations.

Comparison to interleaved ECC: An area, power and latency comparison to the interleaved burst-error detection and correction architecture of Li et al. (2018) will yield numerical values quantifying the trade-offs between the two system architectures and highlight the optimum architecture for a given application space.

This project has provided the design architecture and synthesis validation data to demonstrate the viability of all the above future projects, providing a functional and well-characterised baseline architecture for future research into adaptive burst-error ECC on cache memory systems.

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Appendices

Appendix A: Verilog Module File List

Below is the list of Verilog source files making up the full RTL implementation:

Filename	Description
sixburstandfourburstcache.v	Config 2 - top-level integration for 6B+4B Adaptive. Includes ECC_Cache_System top module and the 6-burst and 4-burst encoder-decoder integration as well as mode manager.
threeburstandfourburstcache.v	Config 3 - top-level integration for 4B+3B Adaptive. Achieved by performing adaptive 3-burst and 4-burst correction.
sixburstcache_v_10_33.22_AM	Config 1 - 6-Burst Only configuration source. Single-mode 6-burst BCH encoder-decoder with cache..
fourburstonlycache_v_10_42.11_AM	Config 0 - 4-Burst Only configuration source. Single-mode 4-burst BCH encoder-decoder with cache.

Table 5.1 Verilog Module File List

Appendix B: Cadence Genus Synthesis Configuration

Common settings used in all synthesis runs are as follows:

Tool: Cadence Genus 21.14-s0821

Host: vlsi11.blr.amrita.edu

Technology library: GPDK045 45 nm, slow.lib

Operating condition: Slow corner (1.08V, 125C)

Clock period: 10ns (100 MHz), clock uncertainty: 100ps

Synthesis flow: syngeneric synmap synopt (medium effort)

Output files: netlist.v, sdc, .sdf, area.rpt, power.rpt, timing.rpt, gates.rpt, qor.rpt

Appendix C: Synthesis Results Raw Data — Netlist Files

The generated gate-level netlists for all four configuration will be submitted as individual files:

Netlist File	Configuration
fourburstonlycache_netlist_v_10_42.11_AM	Config 0 — 4-Burst Only
sixburstcache_netlist.v	Config 1 — 6-Burst Only
sixburstandfourburstcache_netlist.v	Config 2 — 6B+4B Adaptive
threeburstandfourburstcache_netlist.v	Config 3 — 4B+3B Adaptive

Table 5.2 Generated gate-level netlists for all four configurations

Gate level netlists are synthesized and simulated using post-synthesis simulation used for static timing analysis and also gate level power estimation using switching activity information (Cadence Joules or Synopsys Prime Time PX analysis using VCD information).

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